

3. The table shows the electronic structures of atoms of the elements **A–F**.

A–F are not the chemical symbols for the elements.

Element	Electronic structure
A	2
B	2,6
C	2,8,1
D	2,8,7
E	2,8
F	2,8,6

(a) Which of elements **A–F** is found in Group 6 and Period 3 of the Periodic Table?

Explain your choice, referring to electronic structure. [2]

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(b) Which **two** of elements **A–F** are chemically inert?

Explain your choice, referring to electronic structure. [2]

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(c) Element **D** has two isotopes. Isotope **1** has 18 neutrons and isotope **2** has 20 neutrons.

Complete the table by giving the atomic number and mass number of both isotopes. [2]

Isotope	Atomic number	Mass number
1		
2		



- (b) Tick (✓) **two** boxes which correctly describe the change in density and boiling point for the elements across Period 3. [2]

The density of metals and non-metals increases

☐

The boiling point of metals increases but the boiling point of non-metals shows no trend

☐

The density of metals shows no trend but the density of non-metals decreases

☐

The boiling point of metals and non-metals shows no trend

☐

The density of metals increases but the density of non-metals shows no trend

☐

The boiling point of metals shows no trend but the boiling point of non-metals decreases

☐

The density of metals decreases but the density of non-metals shows no trend

☐

- (c) Argon is the next element in Period 3 after chlorine, Cl.

State why it is not possible to predict a melting point for argon using the information in the table. [1]

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(d) Phosphorus is found in phosphoric acid, H_3PO_4 .

(i) During the production of phosphoric acid, phosphorus is heated to 60°C .

Give the state of phosphorus at 60°C . Explain your choice.

[2]

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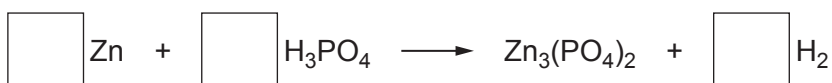
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(ii) Phosphoric acid reacts with zinc to produce zinc phosphate and hydrogen.

Balance the equation for this reaction.

[1]



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